

Chapter 13

One-Factor Experiments: General

13.1 Analysis-of-Variance Technique

In the estimation and hypothesis testing material covered in Chapters 9 and 10, we were restricted in each case to considering no more than two population parameters. Such was the case, for example, in testing for the equality of two population means using independent samples from normal populations with common but unknown variance, where it was necessary to obtain a pooled estimate of σ^2 .

This material dealing in two-sample inference represents a special case of what we call the *one-factor problem*. For example, in Exercise 10.35 on page 357, the survival time was measured for two samples of mice, where one sample received a new serum for leukemia treatment and the other sample received no treatment. In this case, we say that there is *one factor*, namely *treatment*, and the factor is at *two levels*. If several competing treatments were being used in the sampling process, more samples of mice would be necessary. In this case, the problem would involve one factor with more than two levels and thus more than two samples.

In the $k > 2$ sample problem, it will be assumed that there are k samples from k populations. One very common procedure used to deal with testing population means is called the **analysis of variance**, or **ANOVA**.

The analysis of variance is certainly not a new technique to the reader who has followed the material on regression theory. We used the analysis-of-variance approach to partition the total sum of squares into a portion due to regression and a portion due to error.

Suppose in an industrial experiment that an engineer is interested in how the mean absorption of moisture in concrete varies among 5 different concrete aggregates. The samples are exposed to moisture for 48 hours. It is decided that 6 samples are to be tested for each aggregate, requiring a total of 30 samples to be tested. The data are recorded in Table 13.1.

The model for this situation may be set up as follows. There are 6 observations taken from each of 5 populations with means $\mu_1, \mu_2, \dots, \mu_5$, respectively. We may wish to test

$$H_0: \mu_1 = \mu_2 = \dots = \mu_5,$$

$$H_1: \text{At least two of the means are not equal.}$$

Table 13.1: Absorption of Moisture in Concrete Aggregates

Aggregate:	1	2	3	4	5	
	551	595	639	417	563	
	457	580	615	449	631	
	450	508	511	517	522	
	731	583	573	438	613	
	499	633	648	415	656	
	632	517	677	555	679	
Total	3320	3416	3663	2791	3664	16,854
Mean	553.33	569.33	610.50	465.17	610.67	561.80

In addition, we may be interested in making individual comparisons among these 5 population means.

Two Sources of Variability in the Data

In the analysis-of-variance procedure, it is assumed that whatever variation exists among the aggregate averages is attributed to (1) variation in absorption among observations *within* aggregate types and (2) variation *among* aggregate types, that is, due to differences in the chemical composition of the aggregates. The **within-aggregate variation** is, of course, brought about by various causes. Perhaps humidity and temperature conditions were not kept entirely constant throughout the experiment. It is possible that there was a certain amount of heterogeneity in the batches of raw materials that were used. At any rate, we shall consider the within-sample variation to be **chance or random variation**. Part of the goal of the analysis of variance is to determine if the differences among the 5 sample means are what we would expect due to random variation alone or, rather, due to variation beyond merely random effects, i.e., differences in the chemical composition of the aggregates.

Many pointed questions appear at this stage concerning the preceding problem. For example, how many samples must be tested for each aggregate? This is a question that continually haunts the practitioner. In addition, what if the within-sample variation is so large that it is difficult for a statistical procedure to detect the systematic differences? Can we systematically control extraneous sources of variation and thus remove them from the portion we call random variation? We shall attempt to answer these and other questions in the following sections.

13.2 The Strategy of Experimental Design

In Chapters 9 and 10, the notions of estimation and testing for the two-sample case were covered under the important backdrop of the way the experiment is conducted. This falls into the broad category of design of experiments. For example, for the **pooled *t*-test** discussed in Chapter 10, it is assumed that the factor levels (treatments in the mice example) are assigned randomly to the experimental units (mice). The notion of experimental units was discussed in Chapters 9 and 10 and

illustrated through examples. Simply put, experimental units are the units (mice, patients, concrete specimens, time) that **provide the heterogeneity that leads to experimental error** in a scientific investigation. The random assignment eliminates bias that could result with systematic assignment. The goal is to distribute uniformly among the factor levels the risks brought about by the heterogeneity of the experimental units. Random assignment best simulates the conditions that are assumed by the model. In Section 13.7, we discuss **blocking** in experiments. The notion of blocking was presented in Chapters 9 and 10, when comparisons between means were accomplished with **pairing**, that is, the division of the experimental units into homogeneous pairs called **blocks**. The factor levels or treatments are then assigned randomly within blocks. The purpose of blocking is to reduce the effective experimental error. In this chapter, we naturally extend the pairing to larger block sizes, with analysis of variance being the primary analytical tool.

13.3 One-Way Analysis of Variance: Completely Randomized Design (One-Way ANOVA)

Random samples of size n are selected from each of k populations. The k different populations are classified on the basis of a single criterion such as different treatments or groups. Today the term **treatment** is used generally to refer to the various classifications, whether they be different aggregates, different analysts, different fertilizers, or different regions of the country.

Assumptions and Hypotheses in One-Way ANOVA

It is assumed that the k populations are independent and normally distributed with means $\mu_1, \mu_2, \dots, \mu_k$ and common variance σ^2 . As indicated in Section 13.2, these assumptions are made more palatable by randomization. We wish to derive appropriate methods for testing the hypothesis

$$H_0: \mu_1 = \mu_2 = \dots = \mu_k,$$

$$H_1: \text{At least two of the means are not equal.}$$

Let y_{ij} denote the j th observation from the i th treatment and arrange the data as in Table 13.2. Here, $Y_{i.}$ is the total of all observations in the sample from the i th treatment, $\bar{y}_{i.}$ is the mean of all observations in the sample from the i th treatment, $Y_{..}$ is the total of all nk observations, and $\bar{y}_{..}$ is the mean of all nk observations.

Model for One-Way ANOVA

Each observation may be written in the form

$$Y_{ij} = \mu_i + \epsilon_{ij},$$

where ϵ_{ij} measures the deviation of the j th observation of the i th sample from the corresponding treatment mean. The ϵ_{ij} -term represents random error and plays the same role as the error terms in the regression models. An alternative and

Table 13.2: k Random Samples

Treatment:	1	2	...	i	...	k	
	y_{11}	y_{21}	$\cdots$	y_{i1}	$\cdots$	y_{k1}	
	y_{12}	y_{22}	$\cdots$	y_{i2}	$\cdots$	y_{k2}	
	$\vdots$	$\vdots$		$\vdots$		$\vdots$	
	y_{1n}	y_{2n}	$\cdots$	y_{in}	$\cdots$	y_{kn}	
Total	$Y_{1.}$	$Y_{2.}$	$\cdots$	$Y_{i.}$	$\cdots$	$Y_{k.}$	$Y_{..}$
Mean	$\bar{y}_{1.}$	$\bar{y}_{2.}$	$\cdots$	$\bar{y}_{i.}$	$\cdots$	$\bar{y}_{k.}$	$\bar{y}_{..}$

preferred form of this equation is obtained by substituting $\mu_i = \mu + \alpha_i$, subject to the constraint $\sum_{i=1}^k \alpha_i = 0$. Hence, we may write

$$Y_{ij} = \mu + \alpha_i + \epsilon_{ij},$$

where μ is just the **grand mean** of all the μ_i , that is,

$$\mu = \frac{1}{k} \sum_{i=1}^k \mu_i,$$

and α_i is called the **effect** of the i th treatment.

The null hypothesis that the k population means are equal against the alternative that at least two of the means are unequal may now be replaced by the equivalent hypothesis

$$H_0: \alpha_1 = \alpha_2 = \cdots = \alpha_k = 0,$$

$$H_1: \text{At least one of the } \alpha_i \text{ is not equal to zero.}$$

Resolution of Total Variability into Components

Our test will be based on a comparison of two independent estimates of the common population variance σ^2 . These estimates will be obtained by partitioning the total variability of our data, designated by the double summation

$$\sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{..})^2,$$

into two components.

Theorem 13.1: Sum-of-Squares Identity

$$\sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{..})^2 = n \sum_{i=1}^k (\bar{y}_{i.} - \bar{y}_{..})^2 + \sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2$$

It will be convenient in what follows to identify the terms of the sum-of-squares identity by the following notation:

Three Important
Measures of
Variability

$$SST = \sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{..})^2 = \text{total sum of squares,}$$

$$SSA = n \sum_{i=1}^k (\bar{y}_{i.} - \bar{y}_{..})^2 = \text{treatment sum of squares,}$$

$$SSE = \sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2 = \text{error sum of squares.}$$

The sum-of-squares identity can then be represented symbolically by the equation

$$SST = SSA + SSE.$$

The identity above expresses how between-treatment and within-treatment variation add to the total sum of squares. However, much insight can be gained by investigating the **expected value of both SSA and SSE**. Eventually, we shall develop variance estimates that formulate the ratio to be used to test the equality of population means.

Theorem 13.2:

$$E(SSA) = (k-1)\sigma^2 + n \sum_{i=1}^k \alpha_i^2$$

The proof of the theorem is left as an exercise (see Review Exercise 13.53 on page 556).

If H_0 is true, an estimate of σ^2 , based on $k-1$ degrees of freedom, is provided by this expression:

Treatment Mean
Square

$$s_1^2 = \frac{SSA}{k-1}$$

If H_0 is true and thus each α_i in Theorem 13.2 is equal to zero, we see that

$$E\left(\frac{SSA}{k-1}\right) = \sigma^2,$$

and s_1^2 is an unbiased estimate of σ^2 . However, if H_1 is true, we have

$$E\left(\frac{SSA}{k-1}\right) = \sigma^2 + \frac{n}{k-1} \sum_{i=1}^k \alpha_i^2,$$

and s_1^2 estimates σ^2 plus an additional term, which measures variation due to the systematic effects.

A second and independent estimate of σ^2 , based on $k(n-1)$ degrees of freedom, is this familiar formula:

Error Mean
Square

$$s^2 = \frac{SSE}{k(n-1)}$$

It is instructive to point out the importance of the expected values of the mean squares indicated above. In the next section, we discuss the use of an **F-ratio** with the treatment mean square residing in the numerator. It turns out that when H_1 is true, the presence of the condition $E(s_1^2) > E(s^2)$ suggests that the F -ratio be used in the context of a **one-sided upper-tailed test**. That is, when H_1 is true, we would expect the numerator s_1^2 to exceed the denominator.

Use of F -Test in ANOVA

The estimate s^2 is unbiased regardless of the truth or falsity of the null hypothesis (see Review Exercise 13.52 on page 556). It is important to note that the sum-of-squares identity has partitioned not only the total variability of the data, but also the total number of degrees of freedom. That is,

$$nk - 1 = k - 1 + k(n - 1).$$

F -Ratio for Testing Equality of Means

When H_0 is true, the ratio $f = s_1^2/s^2$ is a value of the random variable F having the F -distribution with $k - 1$ and $k(n - 1)$ degrees of freedom (see Theorem 8.8). Since s_1^2 overestimates σ^2 when H_0 is false, we have a one-tailed test with the critical region entirely in the right tail of the distribution.

The null hypothesis H_0 is rejected at the α -level of significance when

$$f > f_\alpha[k - 1, k(n - 1)].$$

Another approach, the P -value approach, suggests that the evidence in favor of or against H_0 is

$$P = P\{f[k - 1, k(n - 1)] > f\}.$$

The computations for an analysis-of-variance problem are usually summarized in tabular form, as shown in Table 13.3.

Table 13.3: Analysis of Variance for the One-Way ANOVA

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	Computed f
Treatments	SSA	$k - 1$	$s_1^2 = \frac{SSA}{k - 1}$	$\frac{s_1^2}{s^2}$
Error	SSE	$k(n - 1)$	$s^2 = \frac{SSE}{k(n - 1)}$	
Total	SST	$kn - 1$		

Example 13.1: Test the hypothesis $\mu_1 = \mu_2 = \cdots = \mu_5$ at the 0.05 level of significance for the data of Table 13.1 on absorption of moisture by various types of cement aggregates.

Solution: The hypotheses are

$H_0: \mu_1 = \mu_2 = \cdots = \mu_5,$
 $H_1: \text{At least two of the means are not equal.}$
 $\alpha = 0.05.$

Critical region: $f > 2.76$ with $v_1 = 4$ and $v_2 = 25$ degrees of freedom. The sum-of-squares computations give

$SST = 209,377, \quad SSA = 85,356,$
 $SSE = 209,377 - 85,356 = 124,021.$

These results and the remaining computations are exhibited in Figure 13.1 in the SAS ANOVA procedure.

The GLM Procedure					
Dependent Variable: moisture					
Source	DF	Squares	Sum of Mean Square	F Value	Pr > F
Model	4	85356.4667	21339.1167	4.30	0.0088
Error	25	124020.3333	4960.8133		
Corrected Total	29	209376.8000			
R-Square	Coeff Var	Root MSE	moisture Mean		
0.407669	12.53703	70.43304	561.8000		
Source	DF	Type I SS	Mean Square	F Value	Pr > F
aggregate	4	85356.46667	21339.11667	4.30	0.0088

Figure 13.1: SAS output for the analysis-of-variance procedure.

Decision: Reject H_0 and conclude that the aggregates do not have the same mean absorption. The P -value for $f = 4.30$ is 0.0088, which is smaller than 0.05. ▮

In addition to the ANOVA, a box plot was constructed for each aggregate. The plots are shown in Figure 13.2. From these plots it is evident that the absorption is not the same for all aggregates. In fact, it appears as if aggregate 4 stands out from the rest. A more formal analysis showing this result will appear in Exercise 13.21 on page 531.

During experimental work, one often loses some of the desired observations. Experimental animals may die, experimental material may be damaged, or human subjects may drop out of a study. The previous analysis for equal sample size will still be valid if we slightly modify the sum of squares formulas. We now assume the k random samples to be of sizes $n_1, n_2, \dots, n_k$, respectively.

Sum of Squares,
Unequal Sample
Sizes

$$SST = \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{..})^2, \quad SSA = \sum_{i=1}^k n_i (\bar{y}_{i.} - \bar{y}_{..})^2, \quad SSE = SST - SSA$$

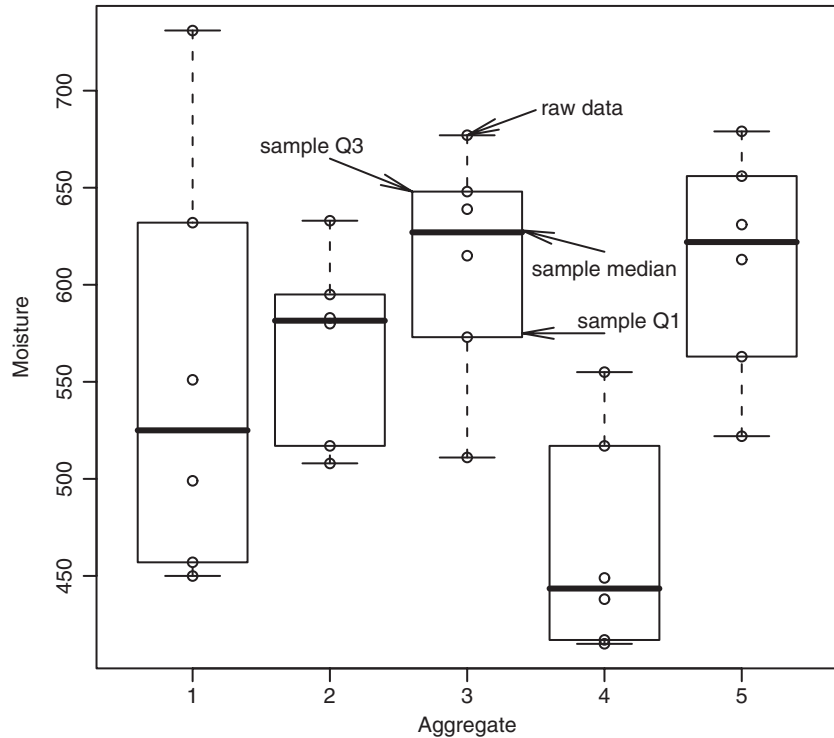


Figure 13.2: Box plots for the absorption of moisture in concrete aggregates.

The degrees of freedom are then partitioned as before: $N - 1$ for SST , $k - 1$ for SSA , and $N - 1 - (k - 1) = N - k$ for SSE , where $N = \sum_{i=1}^k n_i$.

Example 13.2: Part of a study conducted at Virginia Tech was designed to measure serum alkaline phosphatase activity levels (in Bessey-Lowry units) in children with seizure disorders who were receiving anticonvulsant therapy under the care of a private physician. Forty-five subjects were found for the study and categorized into four drug groups:

- G-1: Control (not receiving anticonvulsants and having no history of seizure disorders)
- G-2: Phenobarbital
- G-3: Carbamazepine
- G-4: Other anticonvulsants

From blood samples collected from each subject, the serum alkaline phosphatase activity level was determined and recorded as shown in Table 13.4. Test the hypothesis at the 0.05 level of significance that the average serum alkaline phosphatase activity level is the same for the four drug groups.

Table 13.4: Serum Alkaline Phosphatase Activity Level

G-1		G-2	G-3	G-4
49.20	97.50	97.07	62.10	110.60
44.54	105.00	73.40	94.95	57.10
45.80	58.05	68.50	142.50	117.60
95.84	86.60	91.85	53.00	77.71
30.10	58.35	106.60	175.00	150.00
36.50	72.80	0.57	79.50	82.90
82.30	116.70	0.79	29.50	111.50
87.85	45.15	0.77	78.40	
105.00	70.35	0.81	127.50	
95.22	77.40			

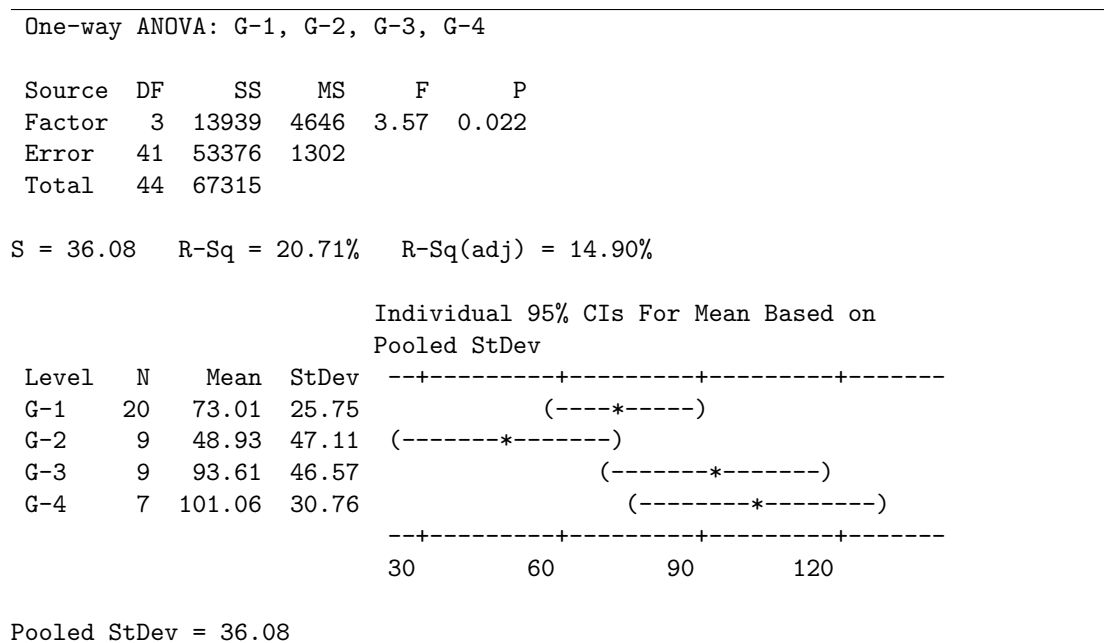
Solution: With the level of significance at 0.05, the hypotheses are

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4,$$

H_1 : At least two of the means are not equal.

Critical region: $f > 2.836$, from interpolating in Table A.6.

Computations: $Y_{1.} = 1460.25$, $Y_{2.} = 440.36$, $Y_{3.} = 842.45$, $Y_{4.} = 707.41$, and $Y_{..} = 3450.47$. The analysis of variance is shown in the *MINITAB* output of Figure 13.3.

Figure 13.3: *MINITAB* analysis of data in Table 13.4.